

QuietKey Gate A Capture Spike

Verdict Report, spike-verdict-01, 16 August 2026

Status: informative input to Gate A1. This spike is one printer, one paper stock, one capture camera. It validates the reader architecture, the mandatory reporting taxonomy pipeline, and produces the first empirical damage envelope. It does not by itself satisfy Gate A1 acceptance, which requires the full print and capture matrix of blueprint QK2-04 section 3.2.

1. Scope and method

29 phone captures: 27 token sheets (2 pristine baselines, 1 dim and 1 glare recapture of the baselines, 23 damage sheets S03 to S25) plus 2 production-format locate pages (S26 pristine, S27 lightly worn). Each token sheet carries 5 tokens of 142 characters: T0 is a never-treated in-sheet control, T1 to T4 received one damage treatment per the campaign log (coffee, water, scuff, fade, scratch, fold, crumple, edge damage, at severities 1 to 3). Ground truth and the frozen v3 reference decoder come from the public repository. A decode success requires RS(83,49) decode plus AEAD tag verification against the per-token Vault Key, so every reported success is cryptographic proof of correct recovery. Silent end-to-end corruption is structurally impossible: an RS miscorrection fails the AEAD tag and the pipeline fails closed. Zero miscorrections occurred in 135 decode attempts.

Reader pipeline as tested: page rectification (homography to 2480x3508), registration of the 15-line token layout against the canonical print geometry, per-line comb fit of character pitch and phase on glyph periodicity with a smooth-in-y robust fit and outlier rejection, per-sheet selection among 6 geometry hypotheses by T0 control accuracy, fixed 19x34 pixel character windows with centroid alignment, nearest-neighbour classification against a blur-augmented baseline glyph bank (honest leave-one-out for baseline sheets, own-T0 exemplar pooling gated on T0 health), confidence thresholding to erasures, RS decode, checksum, AEAD.

2. Operating point: the Gate A reader parameters

Grid search over confidence floor and top-two margin floor on 19,170 recorded character cells. Chosen operating point: `conf_floor` 0.64, `margin_floor` 0.02 (cosine similarity units of this classifier). Any cell below either floor, or with no detectable ink, becomes an erasure. A zero-silent-error operating point does not exist at acceptable decodability: forcing silent errors toward zero (`conf` 0.80, `margin` 0.12) leaves 3 silent errors but collapses decodability from 92 to 34 of 108 treated tokens because erasures explode past the parity budget. Residual confident misreads are the designed load for the 34 parity bytes; they were RS-corrected in every successful decode and are backstopped by checksum and AEAD. These two constants are harness-camera specific and must be re-derived on terminal optics as reproducible-build-controlled values per blueprint section 2.4.

conf	margin	decoded/108	silent errors	erasures
0.55	0.00	88	721	905
0.60	0.01	92	422	1461
0.64	0.02	92	240	1622
0.65	0.01	92	276	1789
0.68	0.02	85	181	~2000
0.80	0.12	34	3	5786

Row 3 is the chosen point. Erasure count for row 5 restated from a neighbouring grid cell; exact per-cell records are in `sweep_records.pkl`.

3. Headline results

Decode success 118 of 135 tokens: 93 of 108 treated tokens, 25 of 27 in-sheet controls, 2 of 2 locate pages. 27 of 27 sheets recovered at least one of their five token positions, including hard crumple, which recovered only its top-of-page control. This directly supports the QK2-04 rule that the footer token is repeated per page: position diversity is what repetition buys.

4. Five-class taxonomy per damage family and severity (treated tokens, 568 characters per cell group)

family	sev	correct	erasure	silent wrong	decoded	RS-corr bytes
baseline	0	1938 (85.3%)	276 (12.1%)	58 (2.6%)	13/16	36
coffee	1	561 (98.8%)	5 (0.9%)	2 (0.4%)	4/4	3
coffee	2	367 (64.6%)	181 (31.9%)	20 (3.5%)	2/4	0

family	sev	correct	erasure	silent wrong	decoded	RS-corr bytes
coffee	3	518 (91.2%)	48 (8.5%)	2 (0.4%)	4/4	3
crumple	1	406 (71.5%)	149 (26.2%)	13 (2.3%)	3/4	7
crumple	2	331 (58.3%)	223 (39.3%)	14 (2.5%)	0/4	0
edge	1	540 (95.1%)	26 (4.6%)	2 (0.4%)	4/4	3
edge	2	561 (98.8%)	6 (1.1%)	1 (0.2%)	4/4	1
edge	3	518 (91.2%)	46 (8.1%)	4 (0.7%)	4/4	1
fade	1	554 (97.5%)	10 (1.8%)	4 (0.7%)	4/4	5
fade	2	565 (99.5%)	3 (0.5%)	0 (0.0%)	4/4	0
fade	3	536 (94.4%)	31 (5.5%)	1 (0.2%)	4/4	1
fold	1	522 (91.9%)	41 (7.2%)	5 (0.9%)	4/4	4
fold	2	544 (95.8%)	21 (3.7%)	3 (0.5%)	4/4	2
fold	3	521 (91.7%)	41 (7.2%)	6 (1.1%)	3/4	3
scratch	1	527 (92.8%)	22 (3.9%)	19 (3.3%)	4/4	10
scratch	2	480 (84.5%)	70 (12.3%)	18 (3.2%)	4/4	6
scratch	3	493 (86.8%)	62 (10.9%)	13 (2.3%)	4/4	7
scuff	1	527 (92.8%)	37 (6.5%)	4 (0.7%)	4/4	3
scuff	2	513 (90.3%)	51 (9.0%)	4 (0.7%)	4/4	2
scuff	3	472 (83.1%)	90 (15.8%)	6 (1.1%)	4/4	3
water	1	507 (89.3%)	55 (9.7%)	6 (1.1%)	4/4	4
water	2	388 (68.3%)	166 (29.2%)	14 (2.5%)	2/4	5
water	3	345 (60.7%)	202 (35.6%)	21 (3.7%)	2/4	5

Reading the table: silent-wrong rates stay between 0.0 and 3.7 percent in every family. Scratch is the worst silent-error family, blade cuts mutate glyphs into other valid characters with confidence, yet all 12 scratch tokens decode because parity absorbs them. Every unrecoverable token fails erasure-dominated: the reader declares unreadability rather than guessing. The baseline row aggregates the four baseline captures including the deliberate dim and glare recaptures.

5. Unrecoverable tokens at the operating point

sheet	token	error bytes	erasure bytes	2E+e (budget 34)
baseline-0-dim-S01	T0	1	35	37
baseline-0-dim-S01	T1	9	29	47
baseline-0-dim-S01	T4	6	50	62
baseline-0-glare-S02	T2	2	42	46
coffee-2-std-S04	T1	7	43	57
coffee-2-std-S04	T4	3	45	51
crumple-1-std-S21	T0	3	37	43
crumple-1-std-S21	T4	0	82	82
crumple-2-std-S22	T1	1	46	48
crumple-2-std-S22	T2	0	43	43
crumple-2-std-S22	T3	4	37	45
crumple-2-std-S22	T4	3	65	71
fold-3-std-S20	T1	3	35	41
water-2-std-S07	T3	1	41	43
water-2-std-S07	T4	2	73	77
water-3-std-S08	T1	8	58	74
water-3-std-S08	T2	3	72	78

The envelope boundary is physical ink destruction plus one capture condition: full coffee rings at severity 2, hard crumple at severity 2, water smear and soak at severities 2 and 3, one borderline fold token at severity 3 (2E+e of 41), and the dim capture, which cost three of five tokens on pristine print. Dim single-frame phone capture is below the envelope; this is a capture-quality problem, not a damage problem, and the production terminal must gate capture quality at acquisition time and use the multi-frame capture the blueprint already recommends.

6. Budget accounting: bytes, not characters

The RS correction condition $2E+e \leq 34$ applies to the 83-byte codeword, not to the 142 printed characters. Each erased 5-bit character erases the one or two codeword bytes it overlaps, so clustered erasures are cheaper than scattered ones, and the 3 sentinel plus 6 checksum characters are outside the codeword entirely. Character-level accounting misled in both directions during analysis. Two consequences for reader conformance: first, the reference decoder frames on the literal sentinel, so an erased sentinel kills an otherwise recoverable token; the sentinel is a public constant and positional repair of it is legitimate structural knowledge, not payload guessing. Second, the printed checksum is consulted only when every character is readable; erased checksum characters cost nothing. With byte-level accounting and sentinel repair, budget arithmetic agreed with the reference decoder on 134 of 135 tokens; the single disagreement was conservative (predicted fail, actual success, adjacent wrong characters sharing a corrupted byte).

7. Locate pages: structural extraction on production-format documents

S26 (pristine): raw read 141 of 142, 1 erasure, decode success. S27 (light wear, fainter print): raw read 142 of 142, decode success. The full production path worked end to end: real recipe document, footer identified structurally as the bottom three wide monospace lines, the 39-character URL prefix and the v-parameter suffix excluded positionally, per-line comb registration, classification, RS, AEAD. Extraction used no URL, domain, body text or genre knowledge, satisfying the section 1.3 independence requirement. Footer glyph pitch on these pages measured 18.7, identical to the token sheets, so scale normalization was a no-op here; the reader nevertheless normalizes pitch and would handle a different footer point size.

8. Reader findings that must carry into the terminal implementation

These four findings came out of failures during the spike and define requirements for the production reader. One: geometry from ink-run endpoints is systematically biased, up to 0.13 pixels per character here, which accumulates to a full character position by line end; character pitch must be estimated from glyph periodicity. Two: paper curl is real and measurable, a sanded sheet showed 12 pixels of x-drift top to bottom after rectification; a single rigid grid cannot fit a curled page and a four-point homography cannot flatten a bulged one; the reader needs per-line registration with a smooth-in-y model and robust outlier rejection, because smeared or stained lines produce confident wrong registrations that must be outvoted by clean lines. Three: this spike selected among geometry hypotheses using the known-content T0

control, which does not exist in production; the terminal must anchor geometry on the sentinel and token structure instead, and this anchoring path needs its own test coverage. Four: erasure quality is the whole game, the confidence signal separates most but not all wrong reads, and the architecture is correct precisely because parity, checksum and AEAD absorb what thresholding cannot.

9. Verdict

GO for the reader architecture and for proceeding toward the formal Gate A1 matrix. The confidence-aware reader with erasure marking, RS(83,49) and AEAD behaved exactly as the blueprint intends: damage becomes erasures, erasures are either within budget or fail loudly, and nothing fails silently. The measured envelope covers every campaign treatment at severity 1, all of edge, fade, fold, scratch and scuff at every severity, and degrades honestly at severity 2 to 3 for treatments that physically remove or relocate ink over large areas. Conditions attached to this GO: the formal Gate A1 run must still execute the full printer, paper and camera matrix; the two reader constants must be re-derived on terminal optics; per-page token repetition remains mandatory; capture-quality gating and multi-frame capture are required terminal behaviours; and the production geometry anchor (sentinel plus structure, no T0) is a named design item requiring test coverage. Nothing in this spike touches or reopens the frozen v3 codec.